Second International Workshop on AI and Serverless (WoAIS) 2026

Part of [27th ACM/IFIP International Middleware Conference (Middleware 2026)](https://2026.middleware-conference.org/) (https://2026.middleware-conference.org/), Universitat Rovira i Virgili, Tarragona, Spain (December 14–18, 2026).

WoAIS2 will be a hybrid workshop with both in-person and remote participants. Please consult the Middleware 2026 organizers for the latest guidance on hybrid format and registration requirements.

The rise of new AI agents (which involve multiple LLM calls, dynamic plans, and a mix of deterministic code and AI models) is creating uniquely complex workloads for serverless platforms. These agentic applications may be triggered by events, run quickly for a few seconds or autonomously for days, and communicate with other agents. This new reality has already resurfaced classic serverless challenges, such as cold starts, state management, and resource allocation, but within the demanding context of modern AI applications. Furthermore, new complications are emerging, including mixed GPU/CPU needs, unpredictable (stochastic) execution plans, long-running yet highly bursty processes, the critical need for robust agent-to-agent communication, and integrations with agentic programming models and frameworks such as LangGraph (https://langchain-ai.github.io/langgraph/), CrewAI (https://www.crewai.com/), AutoGen/AG2 (https://github.com/microsoft/autogen), Google ADK (https://adk.dev/), and IBM's open-source BeeAI (https://www.ibm.com/think/topics/beeai).

Looking beyond the cloud, the scope of serverless is expanding. The workshop looks ahead at future architectures involving AI, hybrid clouds, and especially edge/IoT devices, which current serverless platforms are not well-equipped to support. These next-gen computing architectures are becoming more common but bring new challenges to old concerns such as resource optimization, scaling, cost, monitoring, and ease of use. The serverless experience becomes essential to emerging trends such as DevOps and Platform Engineering (https://platformengineering.org/) in industry and will be critical to the success of next-gen computing.

This naturally leads to a discussion of the role of LLMs and Foundation Models (FMs) in serverless, where the workshop will explore how hybrid serverless platforms can be leveraged for the entire lifecycle of LLMs and FMs (from fine-tuning and customization to inference serving and ongoing management), with a focus on use cases, resource allocations, optimizations, and using AI to improve the serverless experience itself.

A rapidly emerging part of this landscape is the **agent platform layer**: open-source, Kubernetes-native runtimes that support CRD-based agent lifecycle, are framework-agnostic, with native MCP/A2A/OpenTelemetry integrations, with SPIFFE workload identity and Istio Ambient mesh for zero-trust mTLS between agents, as well as enterprise control planes such as IBM watsonx Orchestrate (https://www.ibm.com/products/watsonx-orchestrate) and Amazon Bedrock Agents (https://aws.amazon.com/bedrock/agents/) that unify agents built on different frameworks under a single governance and observability surface. Closely tied to these platforms is the rise of standardized inter-agent protocols. The Model Context Protocol (MCP) (https://modelcontextprotocol.io/) addresses agent-to-tool interaction, while the Agent2Agent (A2A) protocol (https://a2a-protocol.org/latest/) (donated by Google to the Linux Foundation) addresses agent-to-agent discovery, delegation, and coordination. The workshop welcomes work on the systems implications of these platforms and protocols on serverless: zero-trust identity for ephemeral agents, autoscaling for A2A/MCP traffic, cold-start and connection-pooling for protocol endpoints, sandboxed tool execution, and multi-tenant isolation of agents that may originate from many vendors.

A closely related emerging area is the rise of **agentic coding harnesses** such as Claude Code (https://github.com/anthropics/claude-code) and OpenClaw (https://openclaw.ai/), which wrap LLMs with tool use, memory, sandboxed execution, and multi-step planning. These harnesses are simultaneously a representative workload for serverless platforms (bursty multi-LLM sessions, tool calls that fan out to deterministic code and external services, long-lived agent state) and a new programming model that shapes how serverless platforms should expose execution, isolation, and inter-agent communication primitives (see protocols discussed above, such as MCP, A2A, and ACP). The workshop welcomes work on running, scaling, securing, and observing such harnesses on serverless infrastructure, including harness-to-harness (multi-agent) orchestration.

The workshop is also interested in **AI-Driven Research for Systems (ADRS)**, the emerging paradigm articulated in recent work such as [Barbarians at the Gate](https://arxiv.org/abs/2510.06189) (https://arxiv.org/abs/2510.06189) and [Let the Barbarians In](https://arxiv.org/abs/2512.14806) (https://arxiv.org/abs/2512.14806), in which LLM-based agents iteratively generate, evaluate, and refine systems-level solutions and have already matched or outperformed human state-of-the-art in domains such as scheduling, load balancing, and query optimization. ADRS is doubly relevant to WoAIS: its evaluation loops are themselves elastic, embarrassingly bursty workloads naturally suited to serverless execution, and serverless systems are a promising target for ADRS-driven design and autotuning. Submissions on using ADRS-style methodologies to advance serverless and AI systems research, as well as on the infrastructure needed to operate ADRS at scale, are encouraged.

ADRS belongs to a broader, rapidly growing family of **AI-agent-driven research loops** that share an iterative generate–evaluate–refine pattern, spanning evolutionary code-search agents (e.g., [AlphaEvolve](https://deepmind.google/blog/alphaevolve-a-gemini-powered-coding-agent-for-designing-advanced-algorithms/) (https://deepmind.google/blog/alphaevolve-a-gemini-powered-coding-agent-for-designing-advanced-algorithms/)), end-to-end "AI scientist" systems, and evaluation-infrastructure benchmarks. From a systems perspective these are structurally serverless-shaped workloads: many parallel, independently evaluable trials in sandboxed containers with variable resource needs and mixed CPU/GPU usage. They surface questions for WoAIS such as evaluation-as-a-service at FaaS granularity, secure sandboxed trial execution, cost-aware search heuristics, reproducibility for AI-generated designs, and multi-agent orchestration; submissions on any of these directions are welcome.

This workshop brings together researchers and practitioners to discuss their experiences and ideas for future directions in AI and serverless research. We are looking not only for research papers, experience papers, demonstrations, or position papers but also for live presentations of ongoing work, demonstrations, and anything else that may be interesting to the workshop audience.

The latest version of this CFP is available at http://serverlesscomputing.org/woais2/ (http://serverlesscomputing.org/woais2/)

Topics

This workshop solicits papers from both academia and industry on the state of practice and state of the art in AI and serverless computing. Topics of interest include but are not limited to:

- AI Operating System: system-level infrastructure for scalable intelligence, viewing world models as "operating systems" for future AI agents

- Infrastructure for scaling AI: distributed training and evaluation systems, compilers, and data engines that support continual updates
- Infrastructure and network optimizations for AI and serverless applications
- Cloud services for AI: scalable AI cloud services for tasks such as the inspection of large-scale infrastructures
- Operating AI at scale: extending monitoring and adaptation to large-scale production environments with heterogeneous data streams and real-time costs
- Cost per token, energy usage, cost to operate AI agents and models, cost models, pricing models, and economics of AI and serverless
- Using serverless for AI pipelines
- Using serverless for inference serving of AI models and agents
- Autonomous lifecycle management for AI: transitioning AI to operations and managing the infrastructure challenges for AI adoption across its entire lifecycle
- Orchestrating multiple AI agents running as serverless functions, including coordination and synchronization when agents work on sections of the same task
- Memory layers for AI: memory architectures for LLM-based agents, including episodic vs. semantic memory, retrieval mechanisms, and consolidation of interaction-driven experiences
- Cloud AI security: AI-driven cybersecurity risk analysis and secure machine learning for vulnerability assessment of AI and related technologies using serverless and other scalable techniques
- Elastic AI platforms and pay-as-you-go for GPUs with different cost metrics; using AI assist and generative LLMs such as ChatGPT for building, running, and maintaining serverless-like applications
- Agent platform layer: open-source Kubernetes-native runtimes (CRD-based lifecycle, framework-agnostic, MCP/A2A/OpenTelemetry integrations) and enterprise control planes such as IBM watsonx Orchestrate and Amazon Bedrock Agents
- Zero-trust identity for ephemeral agents (SPIFFE workload identity, Istio Ambient mesh mTLS) and multi-tenant isolation of agents from many vendors
- Standardized inter-agent protocols (Model Context Protocol (MCP), Agent2Agent (A2A), and ACP): autoscaling, cold-start, and connection-pooling for protocol endpoints; sandboxed tool execution
- Agentic programming frameworks (LangGraph, CrewAI, AutoGen/AG2, Google ADK, BeeAI) and their interplay with serverless execution, isolation, and inter-agent communication primitives
- Agentic coding harnesses (Claude Code, OpenClaw, etc.): running, scaling, securing, and observing LLM-with-tools harnesses on serverless infrastructure, including harness-to-harness orchestration

- AI-Driven Research for Systems (ADRS) and AI-agent-driven research loops (FunSearch, AlphaEvolve, CodeEvolve, AIDE, AI Scientist, AI co-scientist, AI-Researcher, R&D-Agent): evaluation-as-a-service at FaaS granularity, secure sandboxed trial execution, cost-aware search heuristics, reproducibility for AI-generated designs, and multi-agent orchestration of specialized research roles
- Evaluation infrastructure for agentic research loops (MLE-Bench, MLR-Bench, MLE-STAR, OpenHands, KernelBench): standardized benchmarks and reproducible results for AI-generated designs
- AI-Native Systems and Autonomous Evolution at Machine Speed: governed autonomy with full provenance for AI-driven changes, cheap simulator-based evaluation (e.g., llm-d / llm-d-inference-sim), multi-timescale control loops, and the boundary between runtime adaptation and design-time evolution
- Multi-cloud and hybrid cloud for AI and serverless computing in Edge, Fog, IoT, etc.
- Supporting customization and running user-provided AI models in any environment: Cloud, Edge, Fog, IoT
- Evaluation, benchmarking, and reproducibility: standardized benchmarks and reproducible AI research results, especially when candidate designs are themselves generated and refined by AI agents
- Mixed GPU, accelerators, and CPU workloads for AI agents
- Running AI and serverless applications with stochastic plans (unpredictable execution paths), bursty long-running processes, and inter-process communications
- Developer experience as we transition from "traditional" serverless and FaaS to AI agentic programming
- Developer productivity to build AI serverless code: from local source to observability and maintenance in production
- Serverless data management for AI, Retrieval-Augmented Generation (RAG), vector and graph databases applied to the serverless experience
- AI and serverless for next-gen computing in industry such as Platform Engineering and Internal Developer Platforms
- Low-code and no-code: new programming abstractions for AI and serverless
- Debugging AI serverless applications
- Use cases, experiences with AI and serverless
- DevOps for AI and serverless
- Confidential computing
- Sustainable computing
- Granular computing
- Super-lightweight containers and WebAssembly
- Swarm intelligence
- Other topics related to AI and serverless computing

Important Dates

Paper Submission: **September 17, 2026** (AOE)
Notification of Acceptance: **October 12, 2026**
Final Camera-Ready Manuscript (Hard Deadline): **October 16, 2026**
Author registration deadline: **TBD**
Workshop: **December 14, 2026**
Conference: **December 14–18, 2026**

Papers and Submissions

Paper submissions

Authors are invited to submit original, unpublished research/application papers that are not being considered in another forum.

Submitted manuscripts must have at most **six (6) pages of technical content**, including text, figures, and appendices, but excluding any number of additional pages for bibliographic references. Papers must adhere to the formatting instructions of the ACM SIGPLAN style, which can be found on the ACM template page (https://www.acm.org/publications/proceedings-template); the font size has to be set to 10pt.

Shorter formats such as extended abstracts (5 pages or less), posters, or position papers are also accepted.

Note that submissions must be double-blind: authors' names must not appear on the manuscript, and authors must make a good-faith attempt to anonymize their submissions. References to authors' previous work should be done in the third person to not reveal their identities. There should be no acknowledgments of people or projects. Supplementary material (e.g., GitHub or GitLab repository) should not reveal the authors' identities; to this end, anonymized repositories can be used (e.g., https://anonymous.4open.science (https://anonymous.4open.science)). See below for more details on the anonymity requirements for doubly-anonymous reviewing.

The Middleware conference organizers will provide companion proceedings including all workshop papers, which will be available in the ACM Digital Library, subject to the availability of camera-ready papers by the deadline.

Authors should submit the manuscript in PDF format. All manuscripts will be reviewed and judged on correctness, originality, technical strength, rigour in analysis, quality of results, quality of presentation, and interest and relevance to the conference attendees. Papers will be submitted through **HotCRP**, in a manner consistent with how Middleware workshops use the system; the submission link will be announced on this page and at http://serverlesscomputing.org/woais2/ (http://serverlesscomputing.org/woais2/).

All submitted manuscripts will be peer-reviewed by at least 3 program committee members. Accepted papers with confirmed presentation will appear in the conference proceedings as well as in the ACM Digital Library.

The authors of accepted papers will be given a choice between different copyright agreements, according to the recent changes in the ACM policy. The options will include opportunities for open access as well as the traditional ACM copyright agreement.

Note that at least one author of each accepted workshop paper must hold a full pre-conference registration.

ACM Policies

By submitting your article to an ACM Publication, you are hereby acknowledging that you and your co-authors are subject to all ACM Publications Policies (https://www.acm.org/publications/policies), including ACM's Publications Policy on Research Involving Human Participants and Subjects (https://www.acm.org/publications/policies/research-involving-human-participants-and-subjects). Alleged violations of this policy or any ACM Publications Policy will be investigated by ACM and may result in a full retraction of your paper, in addition to other potential penalties, as per ACM Publications Policy.

Please ensure that you and your co-authors obtain an ORCID ID (https://orcid.org/register), so you can complete the publishing process for your accepted paper. ACM has been involved in ORCID from the start and has recently made a commitment to collect ORCID IDs from all published authors (https://authors.acm.org/author-resources/orcid-faqs) to improve author discoverability, ensure proper attribution, and contribute to ongoing community efforts around name normalization.

ACM Open Access publishing model for 2026 ACM Conferences

Starting January 1, 2026, ACM is fully transitioning to Open Access. All ACM publications, including those from ACM-sponsored conferences, are 100% Open Access. Authors have two primary options for publishing Open Access articles with ACM: the ACM Open institutional model or by paying Article Processing Charges (APCs). With over 2,600 institutions already part of ACM Open, the majority of ACM-sponsored conference papers will not require APCs from authors or conferences (currently, around 76%).

Authors from institutions not participating in ACM Open will need to pay an APC to publish their papers, unless they qualify for a financial waiver. To find out whether an APC applies to your article, please consult the list of participating institutions in ACM Open and review the policy on discretionary open-access APC waivers (https://www.acm.org/publications/policies/policy-on-discretionary-open-access-apc-waivers). Waivers are rare and are granted based on specific criteria set by ACM.

To ease the transition, ACM has approved a temporary subsidy for 2026, offering:

$250 APC for ACM/SIG members

$350 for non-members

This represents a 65% discount (https://www.acm.org/publications/openaccess), funded directly by ACM. Authors are encouraged to advocate for their institutions to join ACM Open during this transition period.

Anonymity Requirements for Doubly-Anonymous Reviewing

Every research paper submitted to WoAIS2 will undergo a "doubly-anonymous" reviewing process: in addition to maintaining the anonymity of the reviewers of the papers, the PC members and reviewers will not know the identity of the authors. To ensure the anonymity of authorship, authors must at least do the following:

1. Authors' names and affiliations must not appear on the title page or elsewhere in the paper.
2. Funding sources must not be acknowledged anywhere in the paper under review; these can be added to accepted papers upon submission of the camera-ready manuscript.
3. Non-anonymized links to the authors' online content must be removed.
4. Research group members, or other colleagues or collaborators, must not be acknowledged anywhere in the paper.
5. The paper's file name must not identify the authors of the paper.

Authors should also use care in referring to related past work. The solution is to reference past work in the third person (in the same way that one would reference work by anyone else). This allows you to set the context for your submission while at the same time preserving anonymity.

Despite the anonymity requirements, authors should still include all relevant work, including their own; omitting them could reveal the author's identity by negation. However, self-references should be limited to the essential ones, and extended versions of the submitted paper (e.g., technical reports or URLs for downloadable versions) must not be referenced. The goal is to preserve anonymity while allowing the reader to grasp the context of the submitted paper fully. It is the responsibility of authors to do their very best to preserve anonymity. Papers that do not follow the guidelines or potentially reveal the author's identity are subject to immediate rejection.

Other submissions

Authors are invited to submit proposals for demos and other presentations that are not papers.

Proposals must be submitted as short abstracts (not longer than one page) in PDF format using the paper submission system, selecting "Other" as submission type.

Accepted presentations will not be part of the conference proceedings but will be part of the workshop agenda with dedicated time for live presentation (with video backup), questions, etc.

Workshop co-chairs

Paul Castro, IBM Research
Pedro García López, Universitat Rovira i Virgili
Vatche Isahagian, IBM Research
Vinod Muthusamy, IBM Research
Aleksander Slominski, IBM Research

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Dennis Gannon, Indiana University & Formerly Microsoft Research
Arno Jacobsen, MSRG (Middleware Systems Research Group)

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- Dennis Gannon, Indiana University & Formerly Microsoft Research
- Eric Rozner, University of Colorado Boulder
- Etienne Rivière, UCLouvain
- Geoffrey Fox, Indiana University
- Gul Agha, University of Illinois at Urbana-Champaign
- Hans-Arno Jacobsen, MSRG (Middleware Systems Research Group)
- Ian Foster, University of Chicago and Argonne National Laboratory
- Josef Spillner, Zurich University of Applied Sciences
- Kyungyong Lee, Kookmin University
- Lucas Nussbaum, LORIA, France
- Maciej Malawski, AGH University of Science and Technology, Poland
- Maciej Pawlik, Academic Computer Centre CYFRONET of the University of Science and Technology in Cracow
- Marc Sánchez Artigas, Universitat Rovira i Virgili
- Per Persson, Ericsson Research
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